



Introduction to Macroeconomics

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Problem Set 5

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Exercise 1

The short-run aggregate supply curve is given by :

$$Y_t = Y_n + \alpha(\pi_t - \pi_t^e)$$

where Y_t is the short-run GDP, Y_n is the long-run GDP, π_t is the observed inflation rate and π_t^e is the expected inflation rate. Let's assume that the Central Bank is able to control directly inflation (π_t) through the money supply and, therefore, it can surprise the agents with a level of inflation that is higher than what agents expect. What is the result of this policy in the short run? Explain why this result is incompatible with money neutrality in the short run.

Exercise 2

Using the AS/AD model (the aggregate supply and aggregate demand model), please answer the following questions :

2.A The economy begins in a long-run equilibrium, represented by point A, where $Y = Y_1$ and $P = P_1$. Suddenly, a wave of optimism overtakes the economy, and households start to consume more. Which one of the following statements is true (with respect to the initial situation, that is, point A) ?

- (a) In the short run, the output will increase and the price will decrease.
- (b) In the long run, the output will be the same and the price will have increased.
- (c) In the short run, only the price will increase.
- (d) None of the above.

2.B The economy begins in a long-run equilibrium, represented by point A, where $Y = Y_1$ and $P = P_1$. Suddenly, there is an increase in the price of oil. Which one of the following statements is true (with respect to the initial situation, that is, point A)?

- (a) There is an episode of stagflation, in which output decreases and price increases. Once the former has decreased and the latter has increased, they will stay at their new values.
- (b) If the government wants the output to go back as soon as possible to its initial value, it must try to make the aggregate demand shift to the left.
- (c) If the government wants the output to go back as soon as possible to its initial value, it must try to make the aggregate demand shift to the right.
- (d) The government cannot do anything about stagflation, because the increase in the price of oil affects the aggregate supply and not the aggregate demand.

2.C The economy begins in a long-run equilibrium, represented by point A, where $Y = Y_1$ and $P = P_1$. Suddenly, part of the capital of the economy is **permanently** destroyed. Which one of the following statements is true on the long run (with respect to the initial situation, that is, point A)?

- (a) In the long run, the output and the price decrease.
- (b) If the government makes the aggregate demand shift to the right, on the long run it can increase the output back again to its initial value.
- (c) If simultaneously there is a wave of pessimism and the households decide to cut back their consumption, the output decreases but we do not know whether the price level increases or decreases.
- (d) If simultaneously there is a wave of pessimism and the households decide to cut back their consumption, the output and the price decrease.

Exercise 3 (Exam June 2013)

Which of the following statements are correct ? :

- I. Expansionary monetary policy shifts the aggregate demand curve to the right and thus it increases income in the long run.
- II. If the marginal propensity to consume equals 0.6 and a country does not import anything, an increase of government expenditures of 1000 increases income by 2500.
- III. If the marginal propensity to consume equals 0.6 and the marginal propensity to import equals 0.1, then an increase of government expenditures of 1000 increases income by 2000.
- IV. Monetary and fiscal policy are automatic stabilizers, because they are used to stimulate production in times of recessions.

- (a) Only statement I is correct.
- (b) Statements I, II and III are correct and IV is wrong.
- (c) Statements II and III are correct and statements I and IV are wrong.
- (d) Only statement IV is correct.

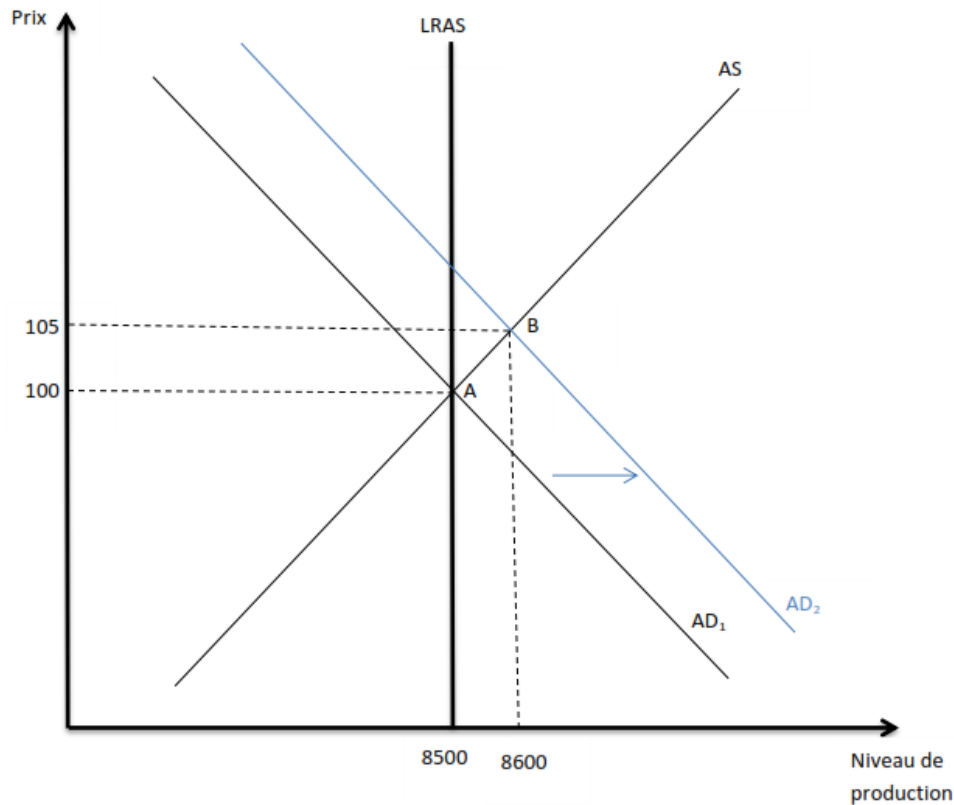
Exercise 4

After World War II, the stabilisation of the economy became an explicit objective of most governments in developed countries. Please answer the following questions :

1. According to some economists, an expansive fiscal policy in times of recession can, under some circumstances, be unable to increase output because consumers fail to increase their expenditures. Why ? Which are the underlying mechanisms ?
2. What are the economic stabilisers ? Why do we call them "stabilizers" and how do they work ?
3. Which are the short-run and long-run advantages and disadvantages of an expansive fiscal policy or an expansive monetary policy in times of recession ?

Exercise 5

Suppose that the Swiss government puts in place a fiscal policy whose effects are represented in the graph below. The level of production is expressed in billions of francs.



1. Discuss the graph (shift in the curve, change in the level of production and prices).
2. Knowing that the production function is given by $Y = 500,000L$ (where Y is the level of production and L is the number of workers) and that the active workforce is equal to 20 million people, calculate the natural unemployment level U_N , the unemployment level U and the inflation rate π .
3. Give the equation of the Phillips curve in the short run. Discuss, and then calculate the value of the slope. Draw the Phillips curve.
4. What is the level of the expected inflation at point A? What would happen if the Swiss National Bank (SNB) were to act so as to maintain the inflation rate at the same level (as in point B) in the years to come? Show graphically (using a Phillips curve) what would happen. Discuss.